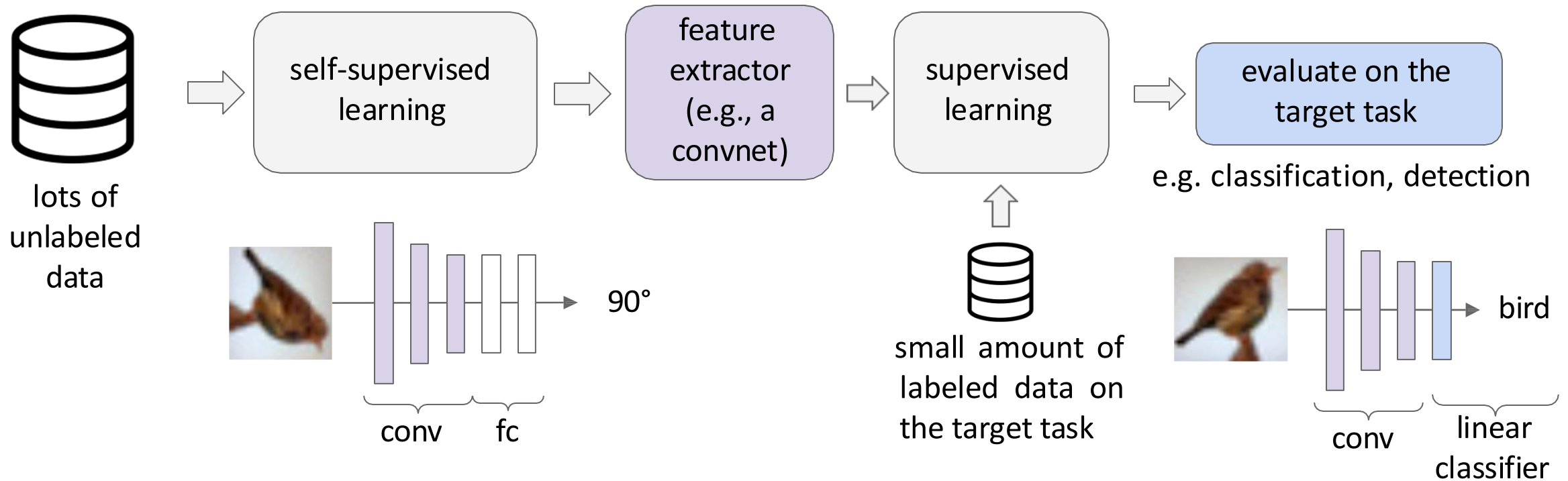


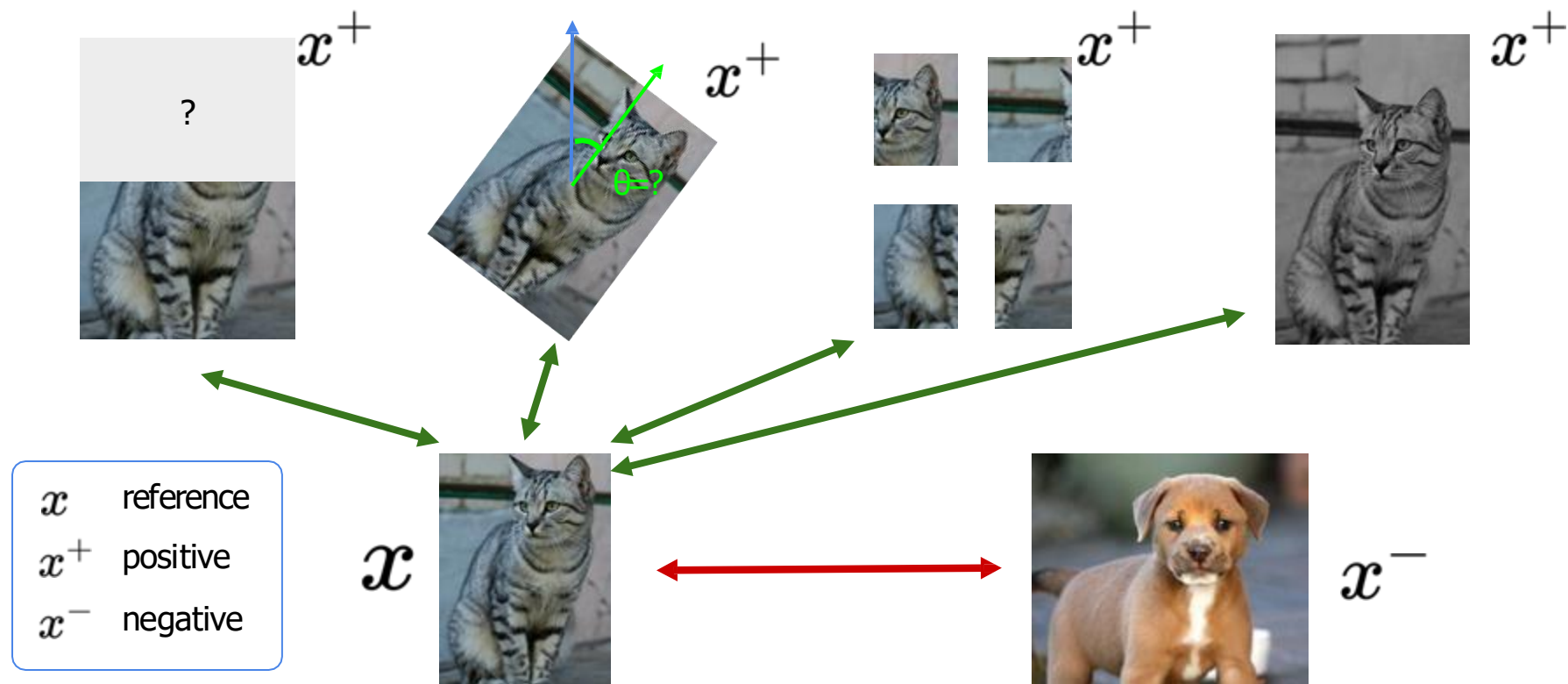
How to evaluate a self-supervised learning method?



1. Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations

2. Attach a shallow network on the feature extractor; train the shallow network on the target task with small amount of labeled data

Contrastive Representation Learning



Contrastive Loss

- Loss given a positive sample and $N - 1$ negative samples:

$$\mathcal{L} = -\mathbb{E} \left[\log \frac{\overset{\text{score for the positive pair}}{\exp \left(s(f(x), f(x^+)) \right)}}{\exp \left(s(f(x), f(x^+)) \right) + \sum_{j \in N-1} \exp \left(s(f(x), f(x^-)) \right)} \right]$$

score for the N-1 negative pairs

- Intuetivly, maximize the positive score from the N samples

SimCLR: A Simple Framework for Contrastive Learning

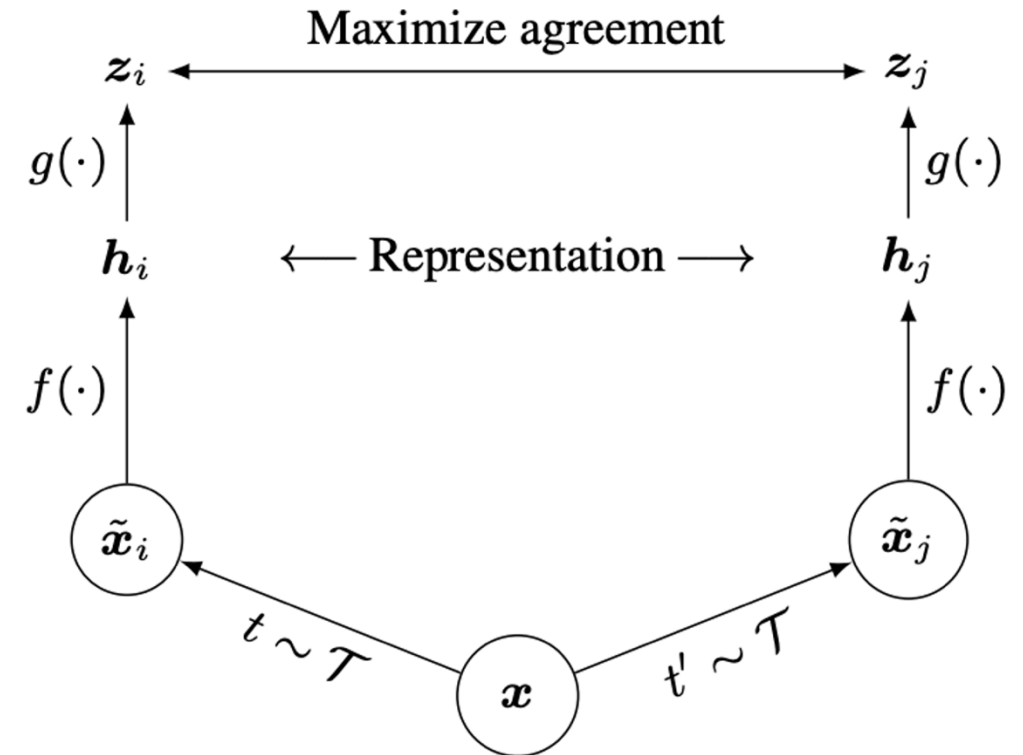
Cosine similarity as the score function:

$$s(u, v) = \frac{u^T v}{||u|| ||v||}$$

Use a projection network $g(\cdot)$ to project features to a space where contrastive learning is applied

Generate positive samples through data augmentation:

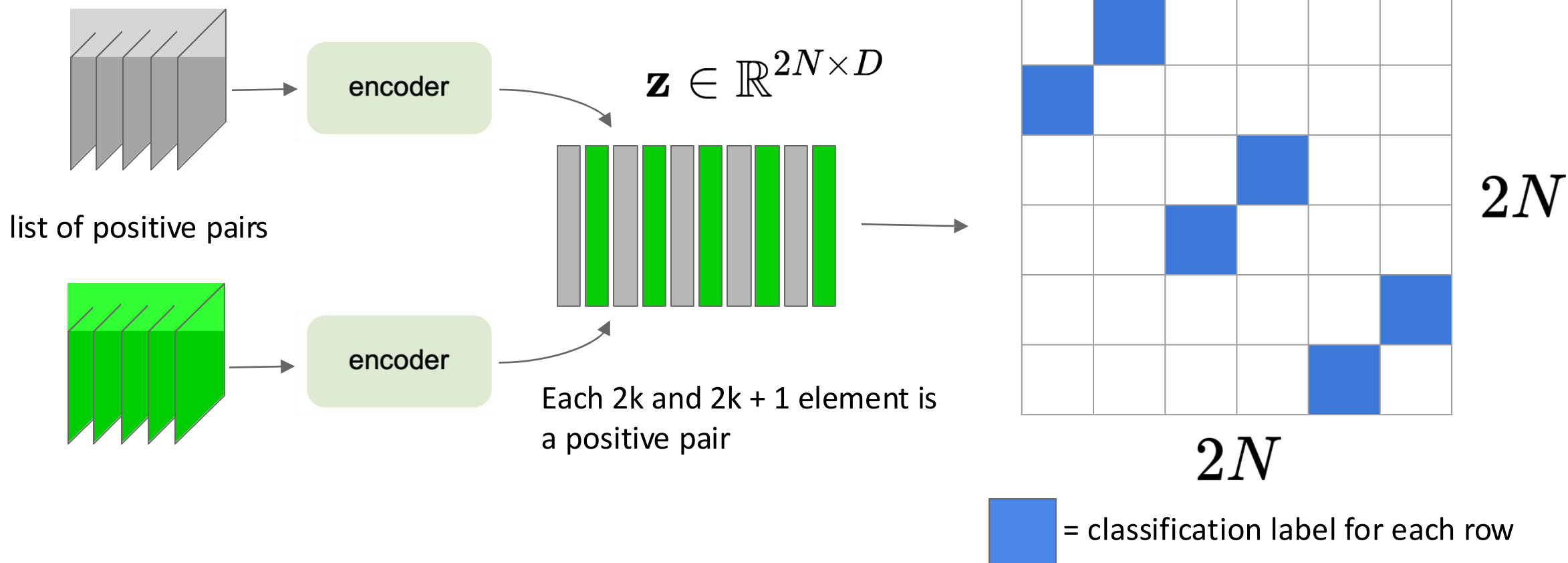
- random cropping, random color distortion, and random blur.



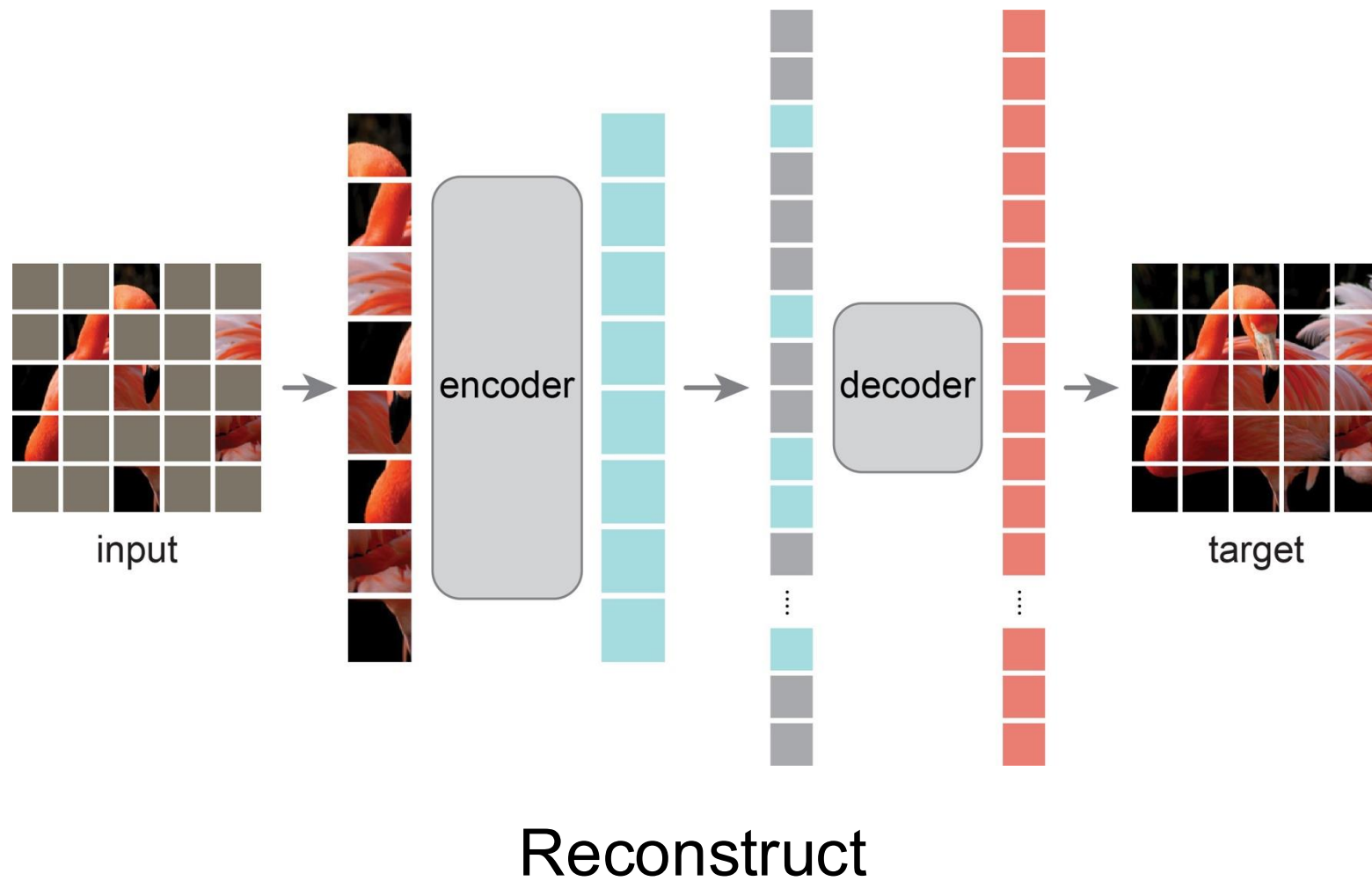
SimCLR: mini-batch training

$$s_{i,j} = \frac{z_i^T z_j}{||z_i|| ||z_j||}$$

"Affinity matrix"



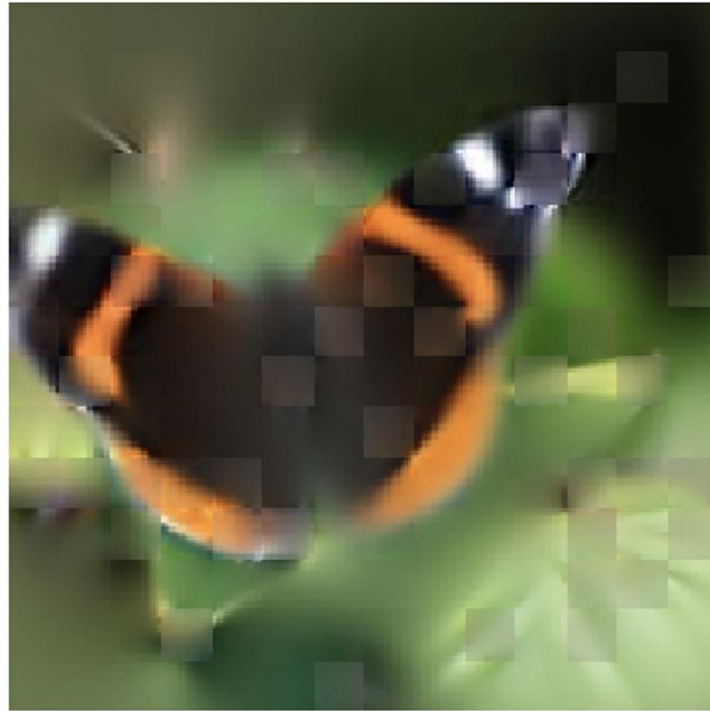
How MAE Works?



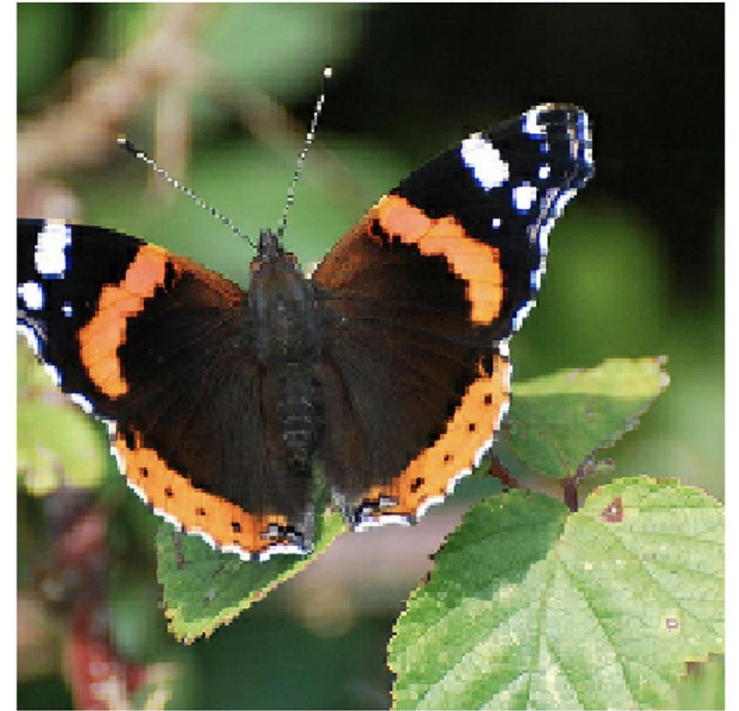
MAE Reconstruction Example



Masked input: 80%



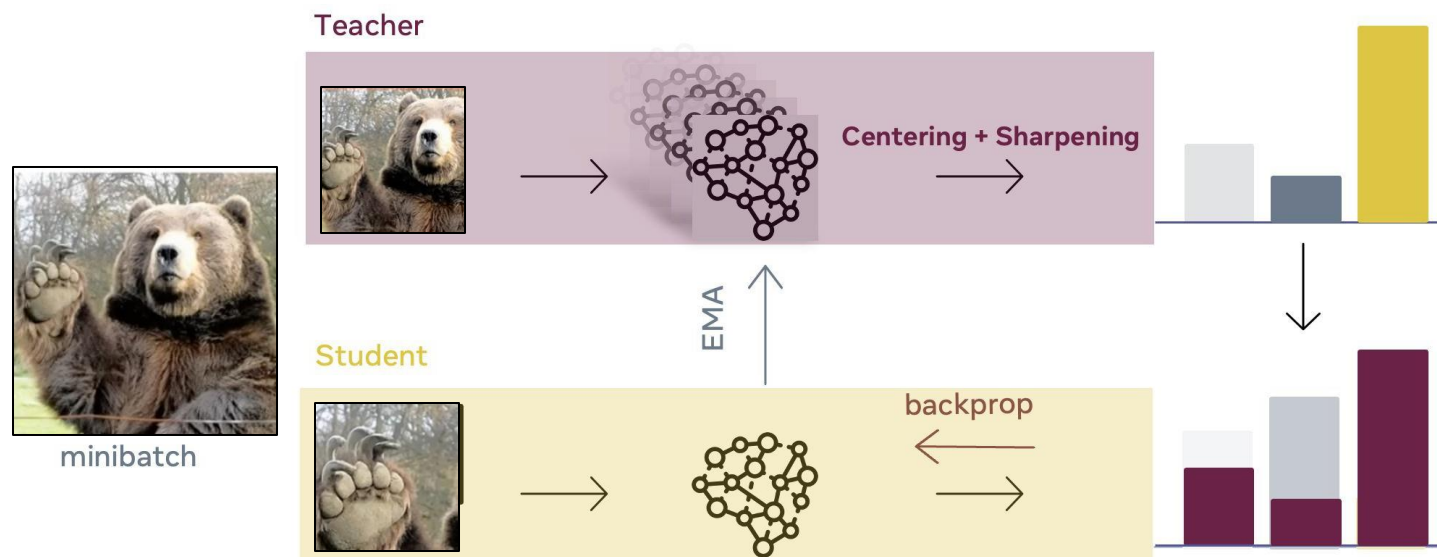
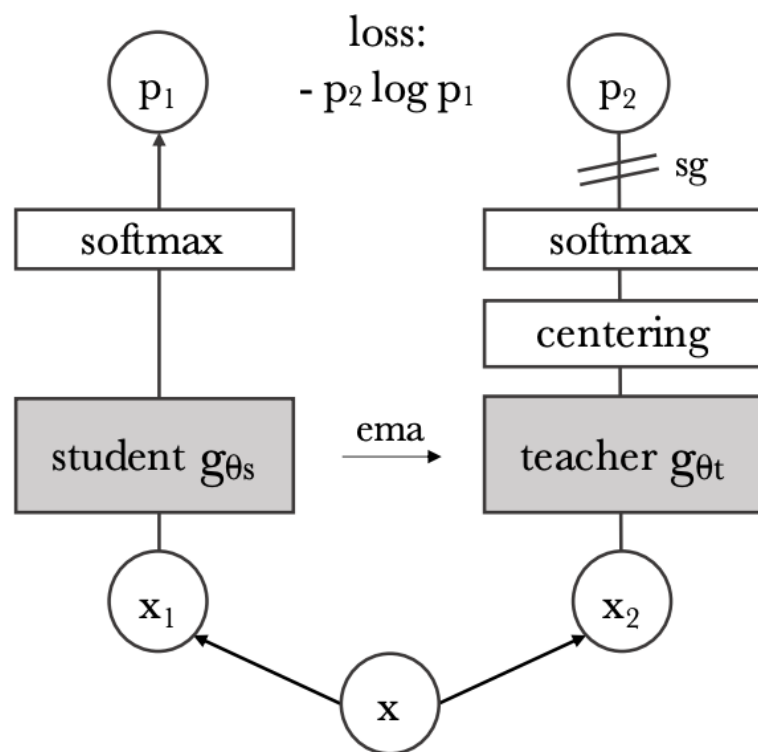
MAE's guess



Ground truth

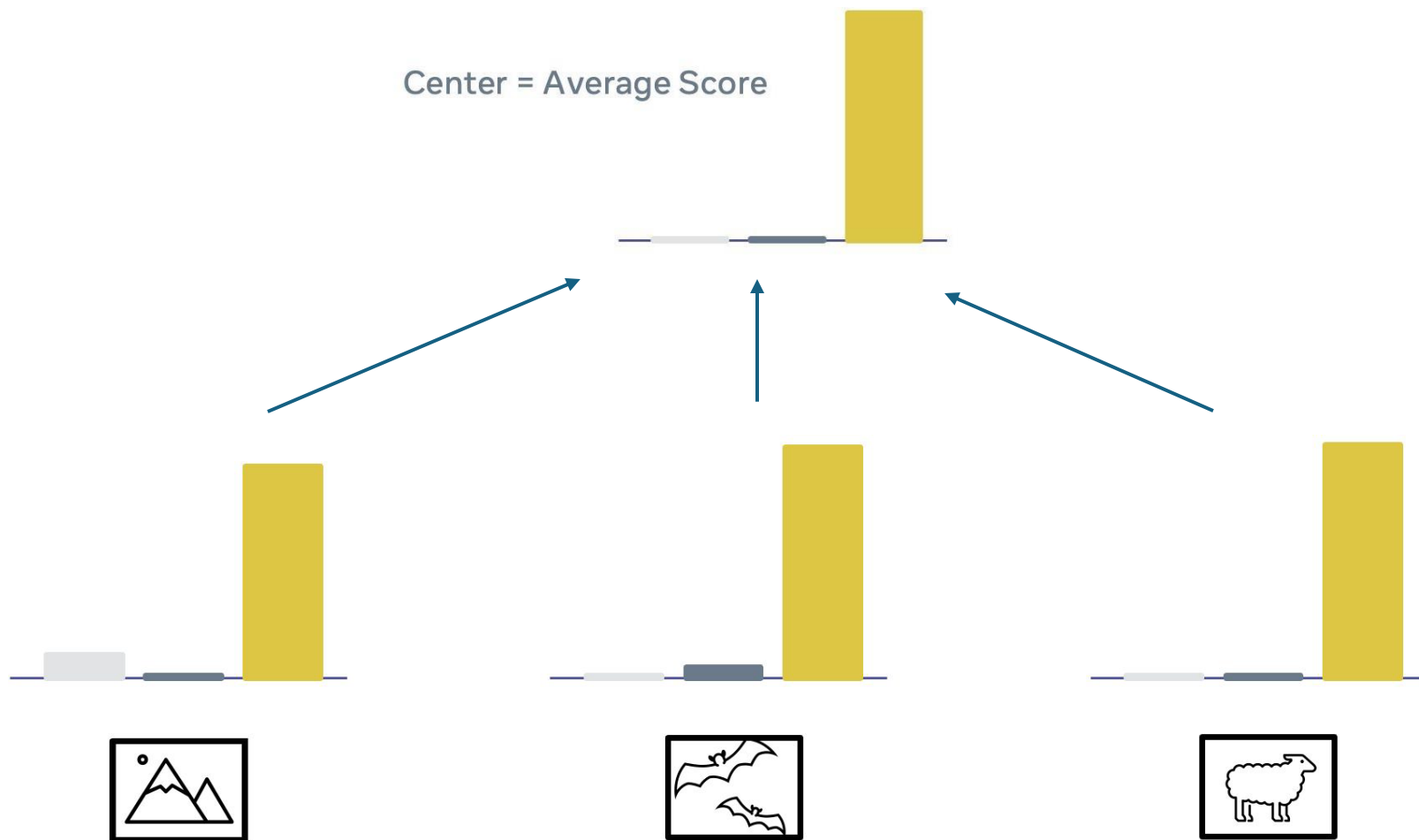
DINO: “Self” Distillation

- DINO learns by matching a “student” logits to a “teacher” logits



$$\theta_{\text{teacher}} \leftarrow \tau \cdot \theta_{\text{teacher}} + (1 - \tau) \cdot \theta_{\text{student}}$$

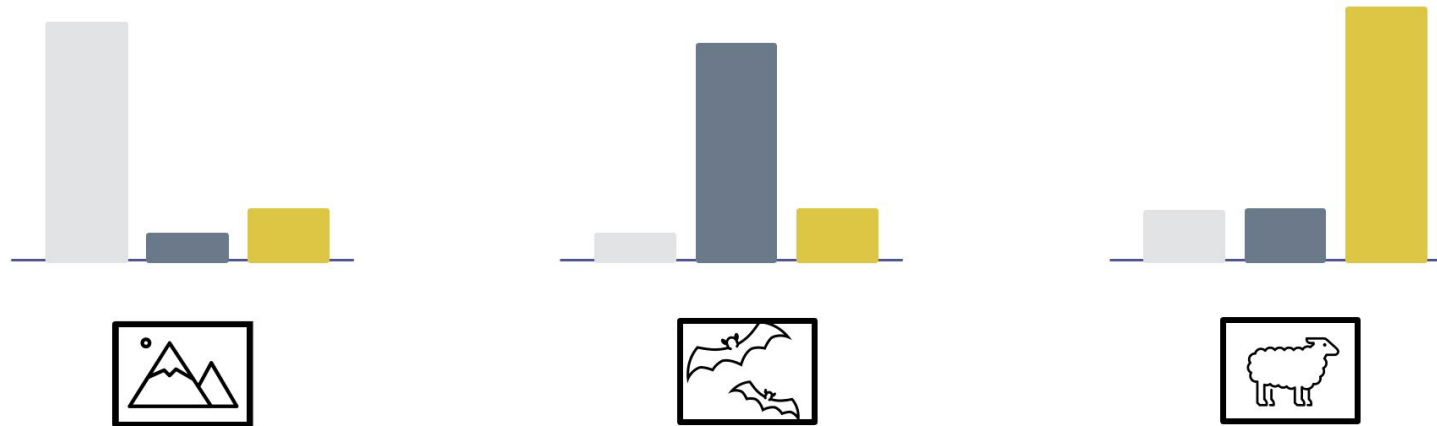
Centering



Sharpening (softmax + temperature)

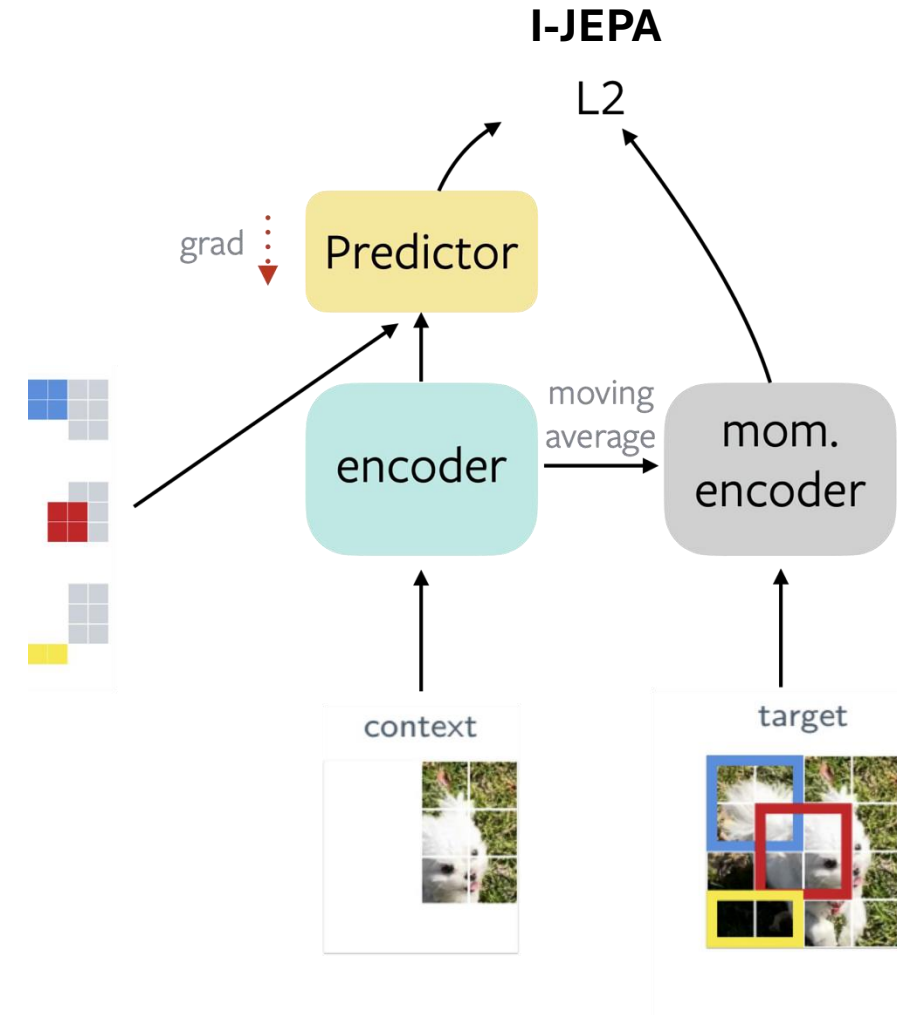
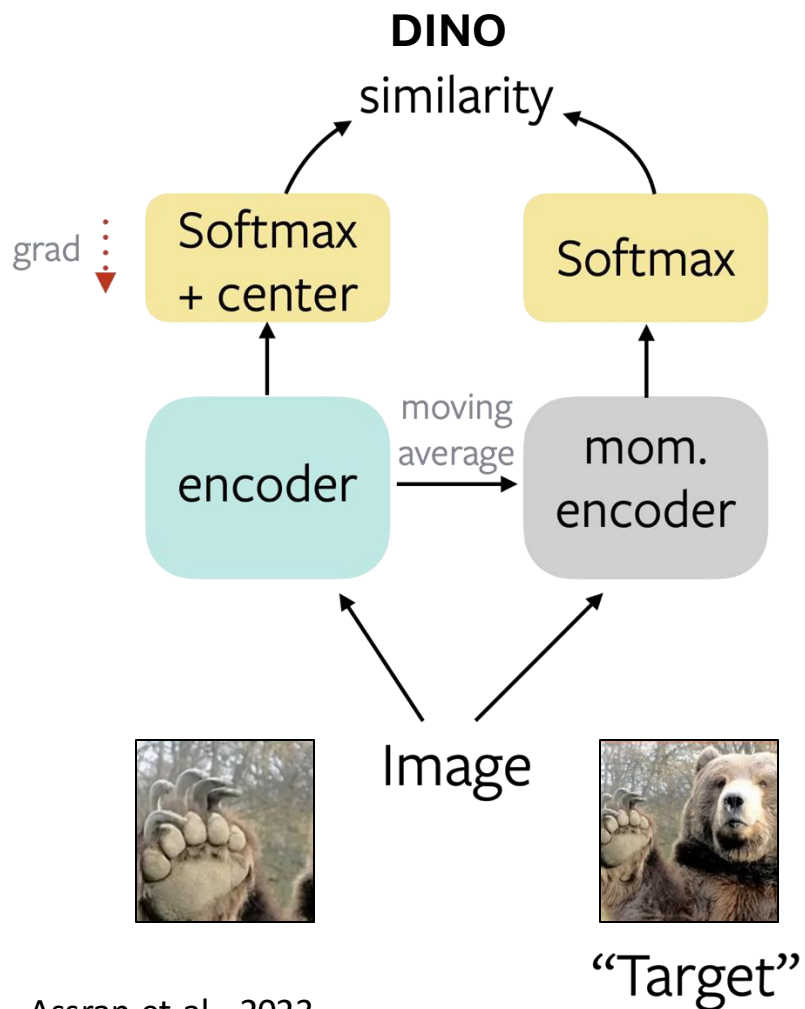
$$p = \text{softmax}\left(\frac{z}{\tau}\right)$$

- Small temperature $\rightarrow$ distribution becomes *sharper* (more peaked)



I-JEPA

I-JEPA: model predicts masked-region **latents** (no softmax targets) conditioned on visible context.

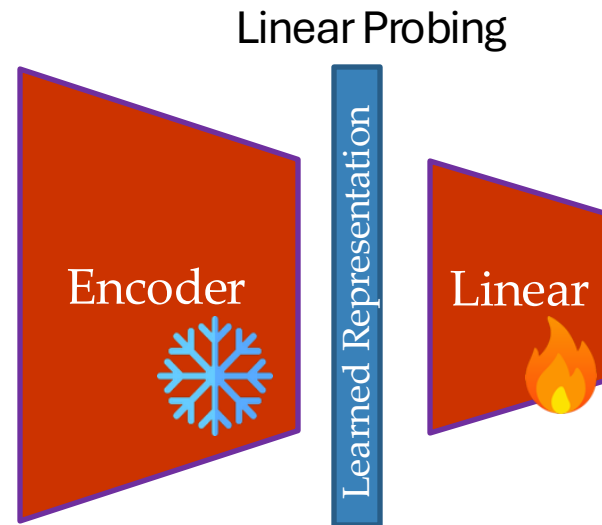
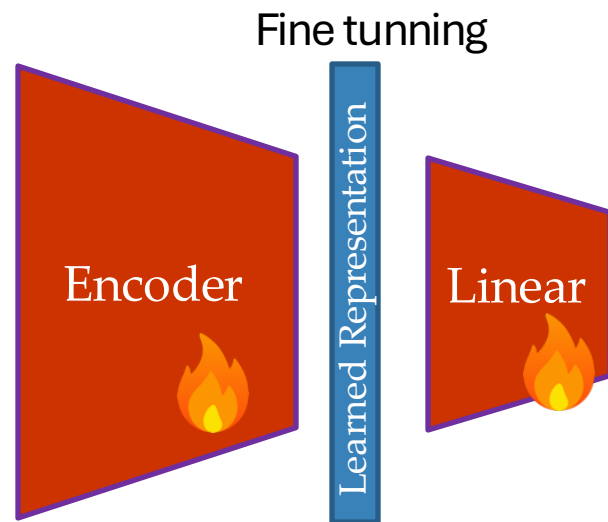


Summarize

- **Contrastive:** SimCLR, CLIP.
- **Generative / Reconstruction:** MAE.
- **Self-distillation:** DINO.
- **Predictive / Conditional:** I-JEPA.

Linear Probing

- In linear probing, the pre-trained model is fixed, and only one linear layer is added
- linear probing: provides a measure of representation quality of a pre-training in restricted conditions



Results

TASK	BENCHMARK	DINO ViT-B/8 0.09B	DINOv2 ViT-G/14 1.1B	DINOv3 ViT-7B/16 7B	SIGLIP 2 ViT-G-OPT/16 1.8B	PE ViT-G/14 1.9B
Segmentation	ADE-20k	31.8	49.5	55.9	42.7	38.9
Depth estimation	NYU ↓	0.537	0.372	0.309	0.494	0.436
Video tracking	DAVIS	68.7	76.6	83.3	62.9	49.8
Instance retrieval	Met	17.1	44.6	55.4	13.9	10.6
Image classification	ImageNet ReaL	85.9	89.9	90.4	90.5	90.4
Image classification	ObjectNet	39.9	66.4	79.0	78.6	80.2
Fine-grained Image classification	iNaturalist 2021	68.3	86.1	89.8	82.7	87.0